Traceable VLA Policy: From Memory to Motion

Training-Free Policy Construction via Retrieval and Modern Hopfield Fusion

Frozen visual and language embeddings retrieve relevant demonstrations. A modern Hopfield memory combines aligned trajectory pieces into robot actions, making the source demonstrations and their contributions inspectable. No additional policy training.

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1 · Toward a Traceable VLA Policy

VLAs such as π0.5 map images, instructions, and robot state to actions. Yet a predicted motion is difficult to trace back to the demonstrations that support it.

Images + text + robot state → **Learned VLA policy** experience in weights → **Robot actions**

LIBERO is a widely used simulation benchmark for VLAs. Its Spatial, Object and Goal suites each test a different ability, and π0.5 succeeds in 98.5% of their episodes. Yet its data allows a **shortcut**. Objects start at nearly the same spot in every demo and most layouts have one task, so the layout alone tells where to reach; a model fit to the demos can overfit to this and ignore the instruction. **LIBERO-PRO** provides more variants of the same tasks, such as swapped object spots or a new instruction, that expose it. For example:

Standard · LIBERO-Object task 2



Salad dressing → basket
π0.5 succeeds: **49/50**

PRO variant 1



Salad dressing → basket
π0.5 grabs the object at the old spot all the time. Succeeds: **0/50** (πRLinf 22/50)

PRO variant 2



Tomato sauce → basket
π0.5 grabs the dressing all the time. Succeeds: **0/50** (πRLinf 2/50)

OUR RESEARCH QUESTION

Can we build every action from retrieved demonstrations, so that it is traceable and does not rely on the data's shortcuts?

2 · Retrieve, associate, and fuse

Proposed traceable policy: frozen features retrieve demonstration pieces.

CURRENT TASK → QUERY q
Image + instruction
Frozen encoders → embeddings

DEMONSTRATION MEMORY
Keys: image / text embeddings
Values: trajectory pieces

Split instruction → one piece per object → **Retrieve** matching demo pieces → **MHN association** weight each memory → **Align + fuse** re-anchor on objects → actions

Object locations + robot state guide alignment.

+ wrist-camera correction (PLANNED)

① Weight each memory piece

$$w_i = \frac{\exp(\beta\, s(q, k_i))}{\Sigma_j \exp(\beta\, s(q, k_j))}$$

② Fuse the re-anchored pieces

$$\hat{\tau} = \sum_i w_i\, \tilde{\tau_i}$$

q query: the current piece's words and look
s feature similarity
$\tilde{\tau_i}$ piece i, re-anchored on the current object
k_i key of memory piece i
β sharpness
$\hat{\tau}$ fused hand path

Fuse compatible pieces after aligning phase and object references; handle gripper events consistently. Grasp and placement may use different demonstrations (4).

3 · Traceable by construction

Which experience?
Retrieved demo pieces and their source tasks

Why this match?
Image / text similarities and association weights

How did it contribute?
Aligned paths and their weighted contributions

Training-free: frozen pretrained models; no additional policy-weight updates. Inputs: RGB, instruction, robot state, plus fixed camera calibration.

Scope: traceable action construction. Embeddings remain learned features; retrieval weights do not establish a causal explanation of task success.

Background: π0.5 (2025) · Modern Hopfield networks (2021) · VINN (2022)

4 · One action, traced to its demonstrations



Query: "Pick up **the salad dressing** and place it in **the basket**"

The instruction is split into two pieces: **pick**, relative to the dressing, and **place**, relative to the basket.

① Which demos?

Pick and place pieces are retrieved on their own, from different demos in memory (6,500 demos of 130 tasks). The 3 most weighted demos of each are shown:

Pick



1 · 4.62% · "...in the tray"
2 · 4.00% · "...in the tray"
3 · 3.25% · "...in the tray"

Place

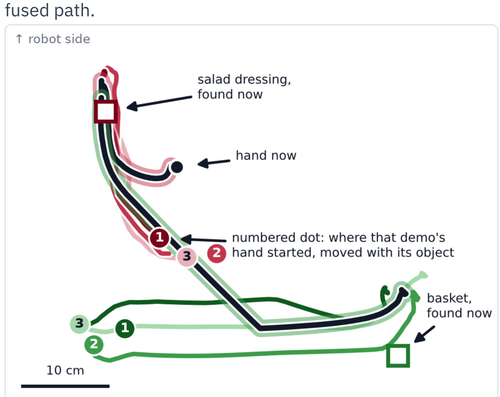


1 · 3.01% · "...in the basket"
2 · 2.88% · "...in the basket"
3 · 2.78% · "...in the basket"

Pick: 1 2 3
Place: 1 2 3
Shade = weight rank: 1 = most weighted

② Line them up

Top view. Each path moves with its own object onto the object found now; pick and place paths mix. Black: the fused path.



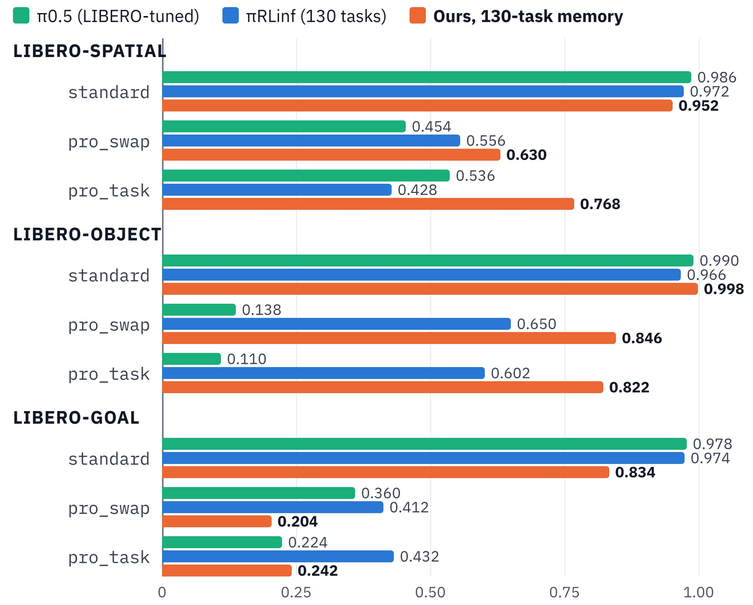
③ Run it here

The fused path in the camera view: to the dressing at its **new** spot (red glow), then into the basket (green glow).



Our method succeeds in 48 of 50 episodes of this case; every step above can be read back from the memory.

5 · Results: success rate per cell



Mean success	9 cells, 30-task mem.	9 cells, 130-task mem.	LIBERO-90	LIBERO-10 + 90
π0.5	0.531	0.531	–	–
πRLinf	0.666	0.666	0.936	0.556
Ours, no wrist step	0.718	0.666	0.817	0.458
Ours	**0.759**	**0.700**	**0.814**	**0.423**

500 episodes per cell (LIBERO-90: 4500). Baselines use no memory, so both 9-cell columns match. Wrist rules were set on episodes 0–9; on unseen episodes 10–49: 0.717 → 0.758.

6 · Where it still fails

- Long horizon: LIBERO-10 0.320 / 0.160 / 0.398 (πRLinf 0.798 / 0.054 / 0.438). On tall objects (moka pot, wine bottle) the wrist step hurts; cause open.
- Goal `pro_task` drops from 0.556 (30-task memory) to 0.242 (130-task); Goal task 3 fails after the grasp in every version.
- Look-alike objects in the wrist view can pull the hand to the twin.

Takeaway. Built from retrieved demonstration pieces, every action traces back to its sources; anchoring the pieces on the objects the instruction names avoids the layout shortcut that VLAs learn, which is where the gains on LIBERO-PRO come from. Long-horizon tasks remain the open gap.